AgriHGT: A Heterogeneous Graph Transformer for Cross-Crop and Cross-Disease Plant Leaf Diagnosis

I. Introduction

Plant diseases remain one of the most persistent threats to global food security, causing substantial reductions in crop yield, produce quality, and farm income in every growing season. Smallholder farmers, who produce a large share of food in regions such as South Asia, often lack timely access to trained plant pathologists, making automated disease diagnosis from leaf images an attractive alternative to manual field inspection. Over the past decade, the availability of large public datasets such as PlantVillage, together with advances in deep learning, has enabled plant disease classification systems to achieve near-perfect accuracy under controlled conditions.

Early approaches relied on convolutional neural networks (CNNs) trained directly on leaf images, sometimes combined with classical image-processing techniques for pre-processing or severity estimation [1]. Deeper CNN architectures, including AlexNet, GoogLeNet, and VGG variants, subsequently achieved classification accuracies exceeding 99% on PlantVillage [2]. More recently, Vision Transformers (ViTs) have exploited global self-attention to improve robustness against symptom variability [3], [4]. Hierarchical transformer architectures such as Swin Transformer have further improved performance on more challenging field datasets [5], while Graph Neural Networks (GNNs) have introduced relational learning by modelling structural dependencies within individual leaf images [6], [7]. Heterogeneous ensemble models that combine segmentation networks, hybrid CNN-transformer classifiers, and metaheuristic optimization have further increased PlantVillage classification accuracy to 99.31% [8].

Despite these advances, existing methods primarily focus on improving visual feature extraction from individual images and rarely exploit relationships among different crops and diseases. Since many plant diseases exhibit similar visual symptoms across different host species, modelling these interrelationships offers an opportunity to improve diagnosis and knowledge transfer beyond single-image classification.

Although CNN-, ViT-, Swin-, and GNN-based approaches have achieved impressive classification performance, nearly all existing methods treat each leaf image as an independent classification instance. Knowledge learned from one disease on a particular crop cannot be explicitly transferred to visually or biologically related diseases affecting other crops. Existing GNN-based methods typically construct homogeneous graphs representing only regions within a single image or focus on one or two crop species [6], [7]. Likewise, heterogeneous ensemble methods improve accuracy through increasingly complex segmentation and optimization pipelines rather than modelling explicit relationships among crops, diseases, and visual characteristics [8]. Consequently, current approaches leave cross-crop and cross-disease relational knowledge largely unexplored.

The primary objective of this study is to develop AgriHGT, a Heterogeneous Graph Transformer capable of exploiting relational knowledge for cross-crop plant disease diagnosis. Specifically, this study aims to:

- Construct a heterogeneous graph that represents crops, diseases, and image-derived visual features as distinct node types.
- Model semantic relationships among these heterogeneous entities using node- and edge-type dependent attention mechanisms.
- Enable knowledge transfer between related diseases occurring on different crop species through heterogeneous message passing.
- Develop an end-to-end plant disease diagnosis framework that captures relational knowledge without relying on complex multi-stage segmentation or ensemble optimization pipelines.

To achieve these objectives, this paper proposes AgriHGT, a Heterogeneous Graph Transformer that explicitly models crops, diseases, and image-derived visual features within a unified heterogeneous graph. The proposed architecture adapts the Heterogeneous Graph Transformer (HGT) framework of Hu et al. [9] to the agricultural domain by introducing node- and edge-type dependent attention over crop–disease–image relationships. Unlike previous methods that classify each image independently, AgriHGT enables relational reasoning and knowledge transfer across different plant species, including Tomato, Papaya, Zucchini, and Bottle Gourd, while avoiding the computational complexity associated with multi-stage segmentation and heterogeneous ensemble frameworks.

The major contributions of this work are summarized as follows:

- A novel heterogeneous graph representation integrating crops, diseases, and image-derived visual features.
- An HGT-based attention mechanism for modelling cross-crop and cross-disease relationships.
- An end-to-end framework capable of transferring disease knowledge across different host plants.
- A relational learning approach that eliminates the need for complex segmentation and ensemble optimization strategies.

II. Related Work

A. CNN-Based Plant Disease Detection

Convolutional neural networks were the first deep architectures to deliver strong results on image-based plant disease classification and remain a common baseline today. Srivastava

et al. [1] combined a CNN classifier with a classical image-processing pipeline on the PlantVillage dataset, reporting 94.87% accuracy, with deployment on IoT devices and drones as the motivating use case. Ferentinos [2] trained several CNN architectures, including AlexNet, GoogLeNet, and VGG variants, on a substantially larger collection of PlantVillage images spanning dozens of plant-disease combinations, achieving 99.53% accuracy on previously unseen test images. Both studies, however, frame disease recognition purely as single-image classification: each leaf is scored independently of every other crop or disease class, so a model that has learned the visual signature of one disease gains no structured benefit from having also seen a related disease on a different host plant, and neither approach generalises naturally beyond the specific crops present in its training data.

B. Vision Transformer and Swin Transformer-Based Approaches

To capture longer-range visual dependencies than the local receptive fields of CNNs allow, several recent works have turned to Vision Transformers (ViTs). Murugavalli and Gopi [3] proposed PLA-ViT, a Precision Leaf Analysis Vision Transformer that combines patch-based self-attention with data augmentation and adaptive learning-rate scheduling, reporting 94.2% accuracy together with favourable inference time and parameter efficiency on PlantVillage. Boukabouya et al. [4] compared several ViT configurations on a PlantVillage-derived tomato-disease subset and reported top accuracies as high as 99.7%, confirming that self-attention-based architectures can match or exceed CNN performance on controlled datasets. Working on the more challenging, field-collected PlantDoc dataset, Liu and Zhang [5] introduced an Efficient Swin Transformer with a Selective Token Generator and a Feature Fusion Aggregator, reducing parameter count by roughly 20.89% relative to standard Swin-T while achieving 80.14% precision and 76.27% recall under realistic, cluttered backgrounds. Parez et al. [10] proposed GreenViT, a fine-tuned Vision Transformer that segments leaf images into sequential patches rather than relying on convolutional pooling, and reported 100%, 98%, and 99% accuracy on the PlantVillage, Data Repository of Leaf Images, and a merged Plant Composite dataset, respectively, outperforming several pretrained CNN baselines. Shandilya et al. [11] combined a CNN branch for fine-grained local features with a ViT branch for long-range contextual dependencies in a hybrid CNN-ViT architecture for maize leaf disease classification, reporting 99.15% accuracy on a combined Kaggle-Mendeley dataset and 95.93% on an independent corn disease-and-severity dataset. Singha Roy and Kukreja [12] trained a ViT with a shared backbone to jointly classify rice leaf disease type and severity level on a custom 3,345-image dataset, achieving a 54.17% weighted-F1 for disease classification and a substantially higher 77.94% weighted-F1 for severity estimation. While these transformer-based methods improve global feature extraction within a single image, none of them model relationships across different crops, diseases, or symptom types; PLA-ViT and the ViT comparison study, in particular, remain confined to the PlantVillage benchmark and a single crop (tomato), respectively, and even the more recent GreenViT, maize CNN-ViT, and rice ViT studies treat every leaf image as an isolated prediction target, with no relational or graph-based reasoning component and no mechanism for transferring symptom knowledge between crops.

C. Graph Neural Network-Based Approaches

A smaller body of work has begun to apply graph neural networks (GNNs) to plant disease imagery, but so far only at the level of structure within a single leaf or a single crop. Ananthajothi and Lokesh [6] construct image-level graphs that connect diseased and healthy leaf regions, with nodes representing leaf parts and edges representing their structural associations, to detect apple scab, cercospora leaf spot, late blight, and powdery mildew on a custom plant image dataset. Elumalai et al. [7] proposed an Attention-Enhanced GNN (AE-GNN) that similarly models leaf images as graphs of discriminative regions, applying an attention-weighting step to refocus on disease-relevant nodes before classifying fifteen fungal diseases (plus healthy classes) on a curated half-spinach and curry-leaf dataset, and reporting that AE-GNN outperforms both CNN and other graph-based baselines. Bera et al. [13] proposed PND-Net, which adds a graph convolutional network on top of a backbone CNN, using region-based and spatial-pyramid pooling to build node features for plant nutrition-deficiency and disease classification, reporting accuracies between 84.30% (PlantDoc) and 96.18% (potato disease) across four datasets. Bajpai et al. [14] fused a GNN branch with a ResNet18 branch for potato leaf disease detection, while Maheswara Rao et al. [15] integrated a CNN with a GCN across 31 disease classes spanning eight crops, in both cases combining visual and graph-structured features within a single crop family rather than across distinct host plants. Surekha and Subha Mastan Rao [16] proposed AdjLeafGNN, which pairs an enhanced CNN (LDDNet) with a GNN to additionally predict the probability of disease spreading between physically adjacent leaves, reporting 98.88% classification accuracy and a 98.71% F1-score on a PlantVillage subset. All of these graph-based approaches demonstrate that relational reasoning can usefully model structure that CNNs and ViTs do not explicitly capture, but their graphs remain homogeneous, image-internal, or restricted to a single crop or a narrow set of one or two species: they provide no mechanism for transferring knowledge between unrelated crops or disease categories.

D. Knowledge-Graph and Hybrid Symbolic-Connectionist Approaches

A parallel line of work has explored knowledge graphs as a way of organising agricultural disease information, though typically from text or alongside, rather than in place of, image-based classification. Li et al. [17] proposed MARBC, a RoBERTa-BiLSTM-CRF model with multi-head attention, to extract entities and relations from a multi-source Chinese rice-disease-and-pest text corpus and construct a heterogeneous knowledge graph, reporting 94.44% F1 on entity-relation

extraction. Zhao et al. [18] combined large language models, an agricultural knowledge graph, and a graph attention network for disease detection in a single crop (*Elaeagnus angustifolia*), reporting 93% accuracy and showing that graph-attention-based fusion outperformed several alternative loss formulations. Lin et al. [19] took a different, purely visual route with GPA-Net, a graph pyramid attention CNN that learns multiscale spatial and graphic relations for fine-grained pest and disease identification, reporting up to 99.0%, 97.0%, and 56.9% accuracy on the cassava-leaf, AI Challenger, and IP102 pest benchmarks, respectively. These studies confirm that graph-structured representations of agricultural knowledge are useful for both textual and visual reasoning, but each remains confined to a single crop or to text-derived entities rather than to image-derived crop, disease, and symptom nodes spanning multiple unrelated host plants.

E. Heterogeneous Ensemble and Hybrid Transformer Approaches

The strongest reported accuracy among the reviewed studies comes from a two-stage heterogeneous ensemble. Kalpana et al. [8] first use U-Net and Swin Transformer V2 to segment diseased leaf regions, then classify the segmented regions using CoAtNet and an enhanced CoAtNet variant, before fusing the predictions of all branches with weights tuned by a Levy Flight Honey Badger Algorithm (LFHBA). On PlantVillage, this pipeline reaches 99.31% accuracy with 99.32% precision and 99.31% F1-score. The result illustrates how far single-image classification accuracy can be pushed with sufficient architectural and optimisation complexity, but it also illustrates the cost of that approach: the pipeline requires a dedicated segmentation stage, two parallel classification branches, and a metaheuristic fusion step, none of which provide any explicit, queryable representation of how the diseases of one crop relate to those of another.

F. Summary and Research Gap

Two patterns emerge consistently across model families, namely CNN, ViT, Swin, GNN, knowledge-graph, and heterogeneous ensemble alike. First, every reviewed method treats disease classification as an independent, per-image (or, for the knowledge-graph studies, per-crop) decision; none constructs an explicit representation of how diseases relate to one another across crops. Second, where relational structure is modelled at all, as in the GNN- and knowledge-graph-based studies [6], [7], [13], [14], [15], [16], [17], [18], [19], it is confined to the interior of a single image, a single crop, or a narrow set of one or two species, rather than spanning the heterogeneous space of crops, diseases, and symptoms that a real agricultural deployment must handle. Pushing single-image accuracy still higher, as the heterogeneous ensemble of Kalpana et al. [8] does, or pushing global visual attention further, as GreenViT, the maize CNN-ViT model, and the rice severity-estimation ViT do [10], [11], [12], comes at the price of substantial architectural and optimisation overhead, without addressing this relational gap at all. AgriHGT is designed to close this gap directly: rather than adding another image-classification branch, it represents crops, diseases, and image-derived features as distinct heterogeneous graph node types and uses a Heterogeneous Graph Transformer to learn attention-weighted relations between them, enabling knowledge transfer across related diseases on entirely different host plants (for example Tomato, Papaya, Zucchini, and Bottle Gourd) within a single, end-to-end model.

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